



VERSION 1.0.0 — MARCH 2026

CodeFusion AI

Software Requirements Specification

Real-Time Collaborative VS Code Extension & Developer Platform

Author	Status	Classification	Last Updated
Engineering Team	Approved	Confidential	March 2026

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1 Project Overview

Purpose

CodeFusion AI is a **VS Code Extension + Web Platform** that enables multiple developers to work on the **same project simultaneously**, similar to Google Docs but built specifically for software development. Unlike traditional Git workflows where developers create branches and merge later, CodeFusion AI allows developers to collaborate in real time while also providing AI assistance, project management, GitHub integration, and team communication.



Real-Time Editing

Multiple developers edit the same files simultaneously with live cursors, selections, and presence detection.



AI-Powered Assistance

Built-in AI project manager, code reviewer, and assistant that helps with tasks, reviews, and documentation.



Integrated Workflow

Seamlessly combines chat, voice, GitHub, task management, and analytics in one platform.

2 Problem Statement

Current development tools have significant limitations that reduce team productivity and increase friction. The following table outlines the key challenges this platform aims to solve:

Challenge	Impact	Solution
Merge Conflicts	Hours of manual resolution, blocked progress	Real-time CRDT-based conflict resolution
Context Switching	Constant tool hopping — Slack, Jira, GitHub, IDE	Unified workspace with all tools integrated
Communication Delays	Waiting for code reviews, async feedback loops	Live collaboration with instant review and AI assistance
Poor Visibility	No real-time view of team progress or blockers	Live dashboards with activity feeds and analytics
No AI Integration	Developers manually handle repetitive tasks	AI project manager, code reviewer, and assistant

3 Objectives

OBJ-01 Enable real-time collaborative coding with sub-50ms latency across distributed teams

OBJ-02 Integrate AI-powered development assistant capable of code generation, review, and explanation

OBJ-03 Provide an integrated project management system with Kanban boards and task tracking

OBJ-04 Implement automatic GitHub synchronization with commit, push, pull, and PR workflows

OBJ-05 Deliver voice and text collaboration tools for seamless team communication

OBJ-06 Provide smart conflict resolution through CRDT-based operational transformation

OBJ-07 Enable live code review with AI-assisted suggestions and automated checks

OBJ-08 Deliver actionable analytics on team productivity, coding time, and project progress

4 Target Users

Software Companies

Enterprise teams needing real-time collaboration and AI assistance

Startups

Small, fast-moving teams that need to ship products quickly

Freelancers

Independent developers collaborating with clients and contractors

Students

Learners working on group projects and pair programming exercises

Open Source Teams

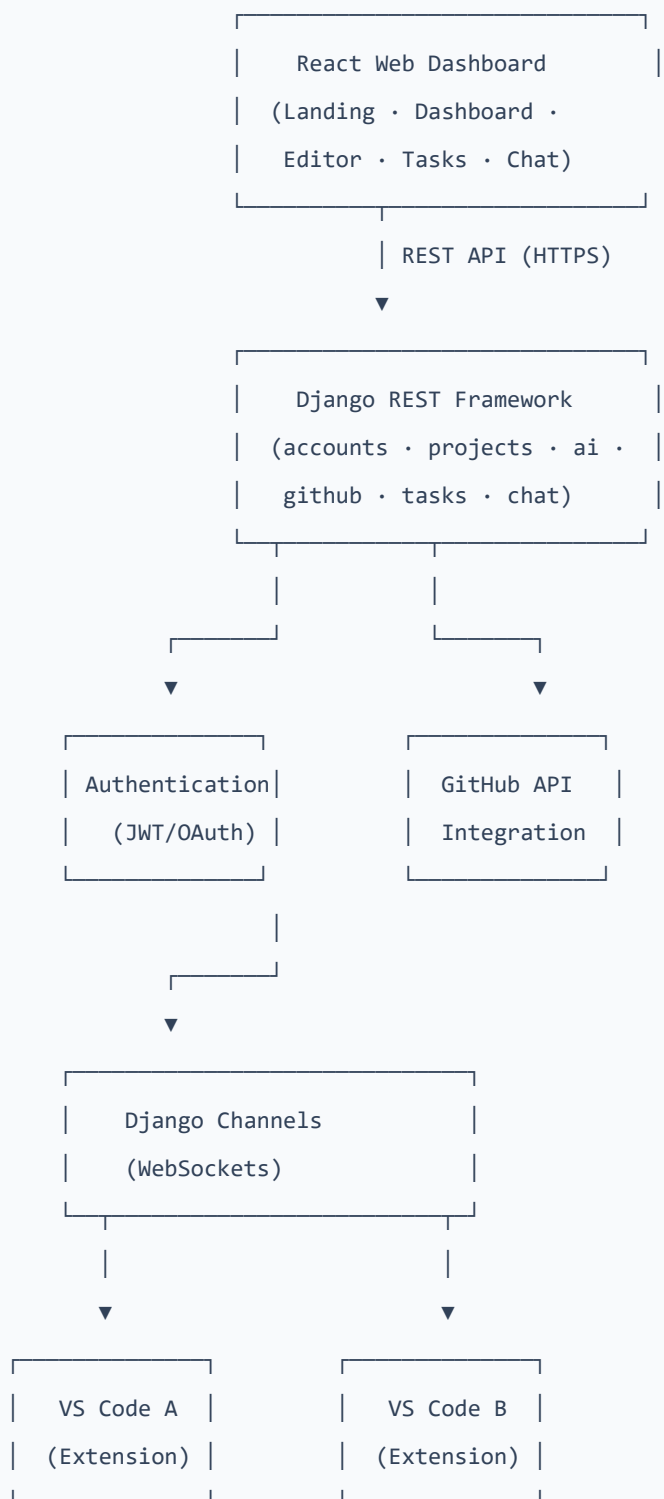
Distributed contributors needing coordinated development

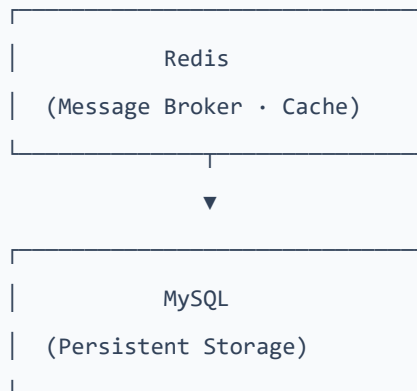
Hackathons

Event organizers and participants needing rapid collaboration

5 System Architecture

The platform employs a multi-tier architecture comprising a React-based web dashboard, Django REST backend with WebSocket support, Redis for message brokering, and MySQL for persistent storage.





6 Technology Stack

Frontend

- React 19
- Vite 7
- Tailwind CSS 4
- Redux Toolkit
- React Router
- Axios

Backend

- Django 5
- DRF
- Django Channels
- Celery
- Redis

Database

- MySQL 8
- Redis (Cache/Broker)

Real-Time Comm.

- WebSockets
- Django Channels
- Redis Pub/Sub
- Yjs (CRDT)

AI / ML

- OpenAI API
- LangChain
- ChromaDB (Vector Store)

DevOps

- Docker
- Docker Compose
- Nginx
- GitHub Actions
- Ubuntu VPS

7 Major Modules

7.1 Authentication

- ✓ Register / Login
- ✓ Google OAuth
- ✓ Forgot Password
- ✓ JWT Token Management
- ✓ GitHub OAuth
- ✓ Email Verification

7.2 Team Management

- ✓ Create Team
- ✓ Role Management
- ✓ Remove Members
- ✓ Invite Members
- ✓ Permission Control

7.3 Projects

- ✓ Create Project
- ✓ Clone Repository
- ✓ Project Settings
- ✓ Import GitHub Repo
- ✓ Invite Developers

7.4 Real-Time Collaboration

- ✓ Live Editing (CRDT)
- ✓ Live Selections
- ✓ Presence Detection
- ✓ Shared File Opening
- ✓ Live Cursors
- ✓ Live Scrolling
- ✓ Typing Indicators
- ✓ Shared Navigation

7.5 Component Ownership

- ✓ Assign Components
- ✓ Request Control
- ✓ Read-Only Mode for Non-Owners
- ✓ File Locking
- ✓ Transfer Ownership

7.6 AI Project Manager



- ✓ Split Tasks
- ✓ Estimate Completion
- ✓ Review Pull Requests
- ✓ Suggest Improvements
- ✓ Assign Components
- ✓ Find Bugs
- ✓ Generate Documentation

7.7 AI Assistant



- ✓ Explain Code
- ✓ Generate Function
- ✓ Fix Bugs
- ✓ Generate Documentation
- ✓ Convert Language
- ✓ Optimize Code
- ✓ Generate Tests
- ✓ Generate README

7.8 Chat System



- ✓ Team Chat
- ✓ File-Specific Chat
- ✓ Mentions
- ✓ Project Chat
- ✓ Thread Replies
- ✓ Emoji Support

7.9 Voice Calls



- ✓ Voice Rooms
- ✓ Mute Controls
- ✓ Screen Sharing
- ✓ Session Recording (Optional)

7.10 GitHub Integration



- ✓ Login
- ✓ Clone
- ✓ Commit
- ✓ Branch Management
- ✓ Pull Request
- ✓ Repository List
- ✓ Pull
- ✓ Push
- ✓ Merge
- ✓ AI Review

7.11 Task Management



- ✓ Kanban Board
- ✓ Task Assignment
- ✓ Priority Labels
- ✓ Todo → In Progress → Testing → Done
- ✓ Due Dates

7.12 Notifications



- ✓ Mentions
- ✓ Commits
- ✓ Build Status
- ✓ Assignments
- ✓ Pull Requests
- ✓ AI Suggestions

7.13 Analytics Dashboard



- ✓ Active Users
- ✓ AI Usage Stats
- ✓ Coding Time
- ✓ Commits Graph
- ✓ Project Progress
- ✓ Productivity Metrics

7.14 File Locking System



- ✓ Lock File
- ✓ Request Control
- ✓ Unlock File
- ✓ Transfer Ownership

7.15 Session Recording



- ✓ Replay Coding Session
- ✓ Activity Logs
- ✓ Timeline View
- ✓ Heat Maps

8 Database Schema

The system employs a relational MySQL database with the following core tables:

Users user authentication and profile data	Teams team grouping and metadata
Projects project configuration and settings	Repositories linked GitHub repositories
Files file metadata and content	Folders directory structure
Commits version history entries	Branches branch metadata
PullRequests PR tracking and reviews	Tasks Kanban task items
Comments code review and chat comments	ChatMessages real-time chat history
VoiceSessions voice call metadata	Notifications user notification queue
Collaborators project membership	Permissions role-based access rules
ActivityLogs audit trail	AISessions AI interaction history
FileLocks locking state per file	CodeChanges operational transform history

9 React Pages

◆ Landing

◆ Login

◆ Register

◆ Dashboard

◆ Projects

◆ Editor

◆ GitHub

◆ Tasks

◆ Calendar

◆ Chat

◆ Voice

◆ Analytics

◆ Notifications

◆ Settings

◆ Profile

◆ Pricing

10 Django Applications

accounts

teams

projects

editor

collaboration

chat

voice

github

tasks

notifications

analytics

ai

billing

api

11 API Specifications

Authentication API

- ↳ POST /api/auth/register
- ↳ POST /api/auth/login
- ↳ POST /api/auth/refresh
- ↳ POST /api/auth/forgot-password

Projects API

- ↳ GET /api/projects
- ↳ POST /api/projects
- ↳ GET /api/projects/:id
- ↳ PATCH /api/projects/:id

Editor API

- ↳ WS /ws/editor/:file_id
- ↳ GET /api/files/:id
- ↳ PUT /api/files/:id

Chat API

- ↳ GET /api/chat/messages
- ↳ POST /api/chat/send
- ↳ WS /ws/chat/:room_id

Voice API

- ↳ POST /api/voice/create-room
- ↳ GET /api/voice/rooms
- ↳ POST /api/voice/join

GitHub API

- ↳ GET /api/github/repos
- ↳ POST /api/github/clone
- ↳ POST /api/github/commit
- ↳ POST /api/github/pr

AI API

- ↳ POST /api/ai/explain
- ↳ POST /api/ai/generate
- ↳ POST /api/ai/review
- ↳ POST /api/ai/optimize

Tasks API

- ↳ GET /api/tasks
- ↳ POST /api/tasks
- ↳ PATCH /api/tasks/:id
- ↳ DELETE /api/tasks/:id

Analytics API

- ↳ GET /api/analytics/dashboard
- ↳ GET /api/analytics/productivity
- ↳ GET /api/analytics/ai-usage

Notification API

- ↳ GET /api/notifications
- ↳ PATCH /api/notifications/:id/read
- ↳ DELETE /api/notifications/:id

12 VS Code Extension Features

✓ Login / Authentication

✓ Join Team

✓ Live Editing

✓ AI Assistant Panel

✓ GitHub Panel

✓ Chat Panel

✓ Voice Panel

✓ Task Panel

✓ Notifications

✓ Activity Feed

✓ File Locking

✓ Presence Indicators

13 Security Architecture

Authentication

- JWT-based token authentication
- OAuth 2.0 (Google, GitHub)
- Email verification
- Session management

Data Protection

- Encrypted WebSocket connections (WSS)
- HTTPS enforcement
- Encrypted secrets management
- Audit logs

Access Control

- Role-Based Access Control (RBAC)
- Project-level permissions
- File-level locks
- Least privilege principle

API Security

- Rate limiting (100 req/min per user)
- Input validation and sanitization
- CORS configuration
- SQL injection prevention

14 Development Roadmap

Phase 1 6–8 Weeks

Core Collaboration Engine

- > VS Code extension scaffold with authentication
- > Team creation and member management
- > Real-time collaborative editing using Yjs (CRDT)
- > Live cursors, selections, and presence detection
- > Basic project management

Phase 2 4–6 Weeks

Communication & Integration

- > Real-time chat system with threads and mentions
- > GitHub integration (clone, commit, push, PR)
- > Kanban task management board
- > File ownership and locking system
- > Notification system

Phase 3 4–6 Weeks

AI-Powered Features

- > AI code assistant (explain, generate, optimize)
- > AI code review for pull requests
- > Automated documentation generation
- > AI-powered code explanations
- > Smart conflict resolution

Phase 4 4 Weeks

Advanced Features & Polish

- > Voice/video collaboration rooms
- > Screen sharing capability
- > Analytics dashboard with productivity metrics
- > Coding session replay and activity logs
- > Enterprise features (SSO, compliance)



Key Technical Challenge

The most critical technical challenge is **real-time synchronization**. The platform must implement a CRDT (Conflict-free Replicated Data Type) library such as **Yjs** or **Automerge**, or an Operational Transformation (OT) engine. These technologies ensure that when multiple developers edit the same file simultaneously, all changes are merged consistently without corrupting the document. This is the foundation upon which the entire real-time collaboration experience is built.

CodeFusion AI

Software Requirements Specification v1.0.0

Confidential Document

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